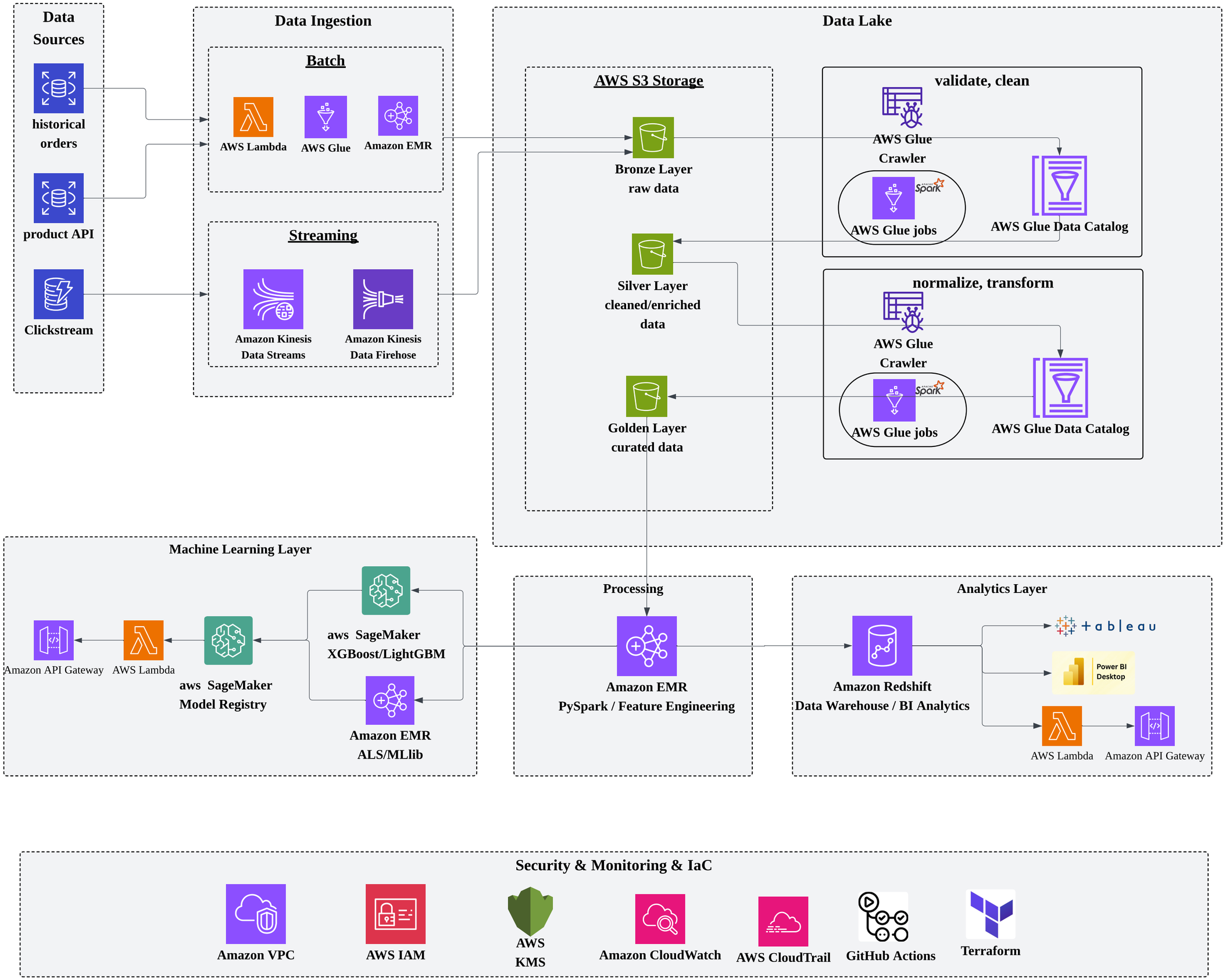
Data Sources
historical orders
product API
Clickstream
Data Ingestion
Batch
AWS Lambda
AWS Glue
Amazon EMR
Streaming
Amazon Kinesis Data Streams
Amazon Kinesis Data Firehose
Data Lake
AWS S3 Storage
Bronze Layer raw data
Silver Layer cleaned/enriched data
Golden Layer curated data
validate, clean
AWS Glue Crawler
AWS Glue jobs
AWS Glue Data Catalog
normalize, transform
AWS Glue Crawler
AWS Glue jobs
AWS Glue Data Catalog
Machine Learning Layer
Amazon API Gateway
AWS Lambda
aws SageMaker Model Registry
aws SageMaker XGBoost/LightGBM
Amazon EMR ALS/MLlib
Processing
Amazon EMR PySpark / Feature Engineering
Analytics Layer
Amazon Redshift Data Warehouse / BI Analytics
tableau
Power BI Desktop
AWS Lambda
Amazon API Gateway
Security & Monitoring & IaC
Amazon VPC
AWS IAM
AWS KMS
Amazon CloudWatch
AWS CloudTrail
GitHub Actions
Terraform